

PROJECT AND TEAM INFORMATION

Project Title

Disaster Resource Allocation and Shelter Management System

Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

In case of disasters like floods and earthquakes , relief shelters register and distribute resources manually and through the use of papers, calls, or even messengers. This makes it possible for certain shelters to be overcrowded while some have capacity left to accommodate more people. Shortages in resources may also not be identified or even tackled, resulting in people in need of immediate aid like the sick and even pregnant women lacking basic supplies that can help them.

State of the Art or Current solution

Current response systems often rely heavily on manual processes , spreadsheets, phone calls, and fragmented platforms to manage shelters and track essential resources. These methods provide limited real-time visibility, making it difficult for authorities to respond efficiently when large numbers of people arrive at shelters simultaneously. While some advanced disaster management systems offer centralized monitoring, they can be expensive, complex , and difficult to deploy for smaller organizations and local authorities. Our solution bridges this gap by providing a centralized, real-time platform for shelter and resource management . It enables authorities to monitor shelter occupancy and resource availability , prioritize assistance based on urgency and vulnerability, and intelligently route affected people to suitable nearby shelters.

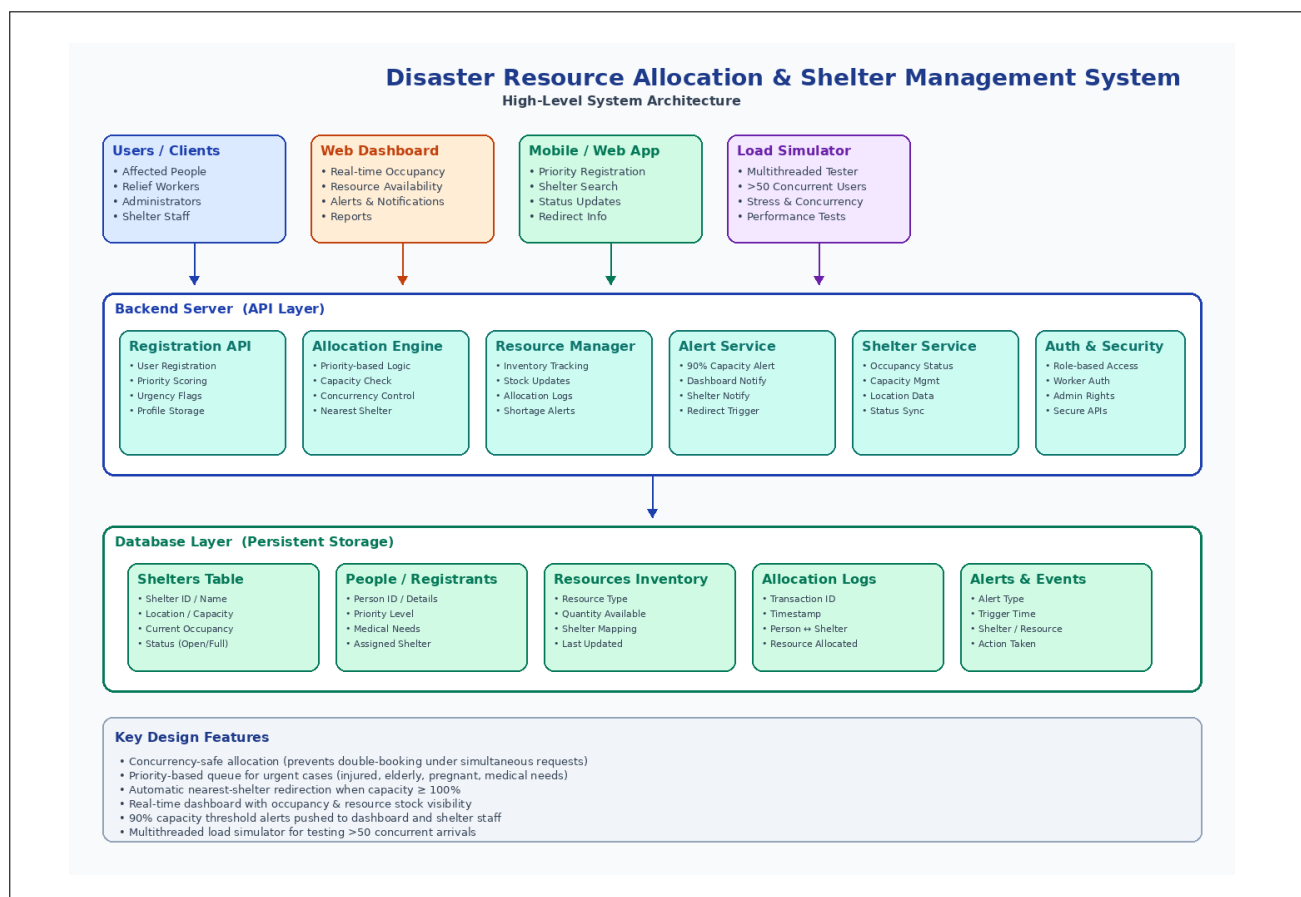
Project Goals and Milestones

- 1. Centralize the disaster management system to store and manage information of the shelters, people, and available resources.*
- 2. Implement live monitoring for relief administrators to track bed occupancy and essential resource levels across all active shelters.*
- 3. Build a triage-based intake engine prioritizing high risk individuals including injured evacuees, senior citizen and pregnant women.*
- 4. Prevent race conditions and duplicate check ins during peak arrival surges using robust database transaction locking.*
- 5. Implement automatic fallback routing to guide individuals to the nearest alternative facility when a target shelter reaches maximum capacity.*

Project Approach

1. *Backend and API layer: Build a centralized REST backend responsible for processing registrations and managing supply allocations and triggering threshold alerts.*
2. *Relational database design: Structure a normalized database schema to track shelter capacities, registrant profiles, supply inventory and transactional logs.*
3. *Priority Scoring: Implement a rule based algorithm that evaluates medical urgency and vulnerability flags to prioritize bed assignments.*
4. *Concurrency Management: Employ ACID compliant database transactions and locking mechanisms to strictly prevent overbooking during simultaneous check ins.*
5. *Administrative Dashboard: Develop an intuitive web interface for real time visualization of occupancy metrics, stock depletion and critical alerts.*

System Architecture (High Level Diagram)(2 pts)



Project Outcome

- A fully functional web based disaster shelter management platform with role based access for relief workers and administrators.
- An interactive real time dashboard displaying shelter occupancy rates,remaining supplies and incoming arrivals.
- An automated triage queue ensuring prompt shelter allocations for high priority and critically vulnerable individuals.
- A multi threaded load testing test bench capable for simulating more than 50 concurrent check in request to validate system stability.
- Dynamic spillover routing that automatically redirects individuals to neighboring shelters upon reaching capacity.

Assumptions

- 1.Shelters will have access to a stable network connection and that will authorize relief workers and administrators to update shelter occupancy and resource information.
2. The deployment environment provided multi core processing and database concurrency handling suitable for simultaneous input output workloads.

References

1. Altay, N. and Green, W. G. (2006). OR/MS Research in Disaster Operations Management. *European Journal of Operational Research*, 175(1), 475–493.
2. Vanajakumari, M., Kumar, S. A., and others (2012). Humanitarian Logistics and Disaster Management. *International Conference on Industrial Engineering and Operations Management*.
3. Caunhye, A. M., Nie, X., and Pokharel, S. (2012). Optimization Models in Emergency Logistics: A Literature Review. *Socio-Economic Planning Sciences*, 46(1), 4–13.